EW Customized Tutor

English, Economics, Politics, and Philosophy

Unit 2 Topics in Business
Lesson 2.4: Determination of Interest and Rationality (Part II)

Pertinent Topics:

- Theories of Rationality
- Some mathematics for economics

Success Criteria:

- Understanding the conceptualizations of Pareto Optimum, Nash, and Expected Value
- Understand the basics of first-degree equations with one unknown

MINDS ON

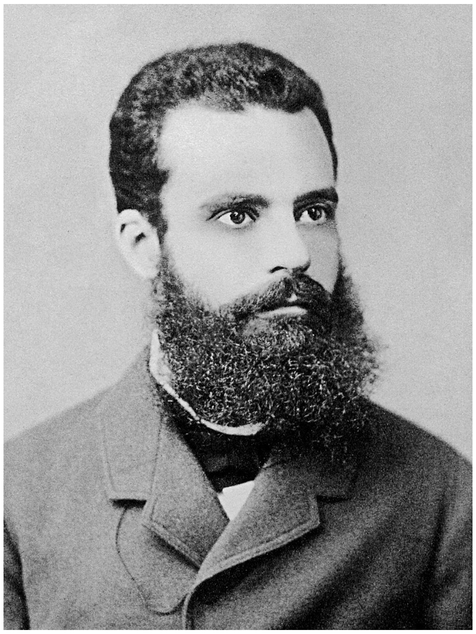
Let’s analyze the ‘game’ you designed this week: “A and B have half of 500,000. Both A and B are separated in two rooms and in front of them are two buttons, (get and give) if both presses get, they both lose half of their money. If they both press ‘give,’ they get to keep their money. If one of them press ‘give’ and one of them press ‘get;’ the ‘get’ person will get 1/3 of the other person’s money.

See if we can construct your scenario into a ‘matrix.’

	Player A		
Actions	give	get	
give	( )	( )	Player B
get	( )	( )	

Pareto Pinciple example: say we are in a group of 10 people and we have to do an assignment; what happens according to Pareto is that 2 people will be doing 80% of the assignment whereas the remaining 8 people will only do 20% of the work.

I. Theories of Rationality



1. Vilfredo Pareto (1848-1923): Pareto is famous for his socioeconomic theories. Particularly, the 80/20 principle. The 80/20 principle is a statistic conclusion that states the counterintuitive socioeconomic dynamics amongst classes, or people:

20% of the population contributes to 80% of the outcomes; or, 20% of the population possess 80% of the resources.

With the context being said, Pareto also introduces an idea called Pareto Optimum: An economy is said to be in a Pareto optimum state when no economic changes can make one individual better off **without making at least one other individual worse off.** In other words, with application of the concepts we learned, e.g., Production Possibilities Frontier.[1]

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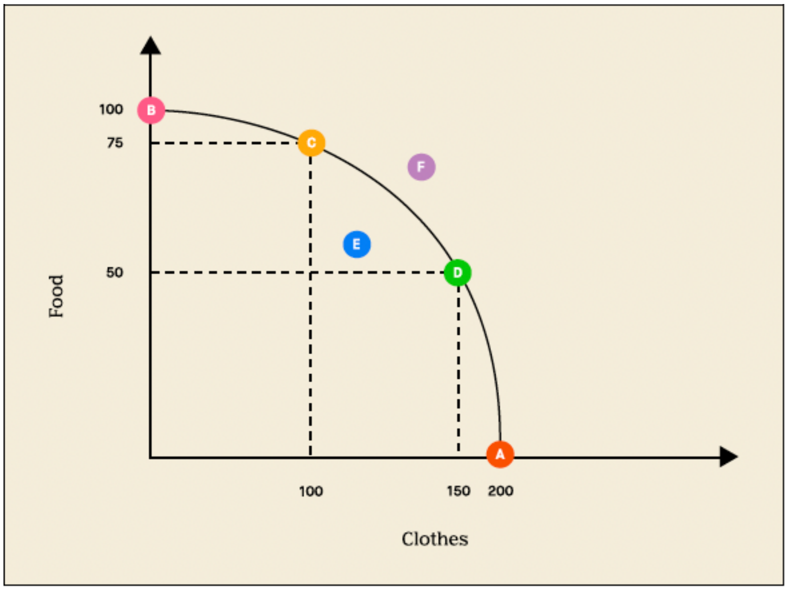
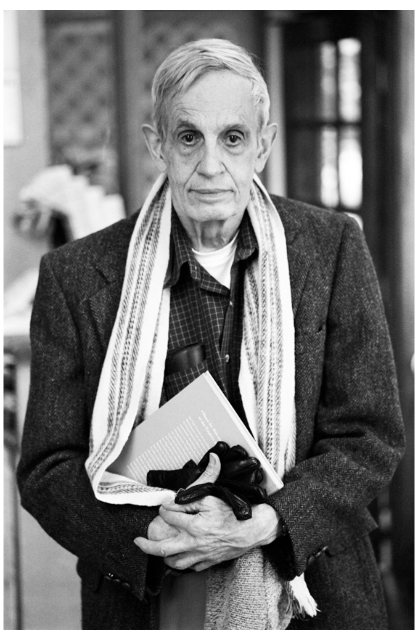
[1] Production Possibilities Frontier measures the 'potential' output of at least two commodities. For example, wheat and apple. To produce 8 apples, you would 'sacrifice' the opportunity cost of producing 4 wheats. And so say for

We call points A, B, C, and Don the frontier, i.e., attainable AND efficient. These four points are Pareto Efficient.



A beautiful mind



2. John Nash (1928-2015): Nash is a mathematician and economist. Building on the works of John von Neumann in game theory, John proposes the famous Nash Equilibrium: a stable state of a system involving the interaction of different participants, in which **no participant can gain by a unilateral change of strategy if the strategies of the others remain unchanged.**

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example, when we produce 2 wheat and 6 apples, out opportunity cost of producing wheat is -2 apples; our opportunity cost of producing apple is -1/2 wheat. (to see the reasonings/calculations: let y be apple and x be wheat we compute: y=-1/2x^2 + 8 for x belongs to [0, 4]) or utility

3. Expected Value: multiplying the **value (utility) of each of the possible outcomes by the likelihood** that each outcome will occur and then summing all of those values. It is a common strategy used for determining whether a game is fair (say a board game, etc.) or for decision making process. For example, let's look at the case for a dice.

General Formula:

Outcome	x_1	x_2	x_3	x_4	...	x_n
Probability	p_1	p_2	p_3	p_4	...	p_n

$$\text{Expected Value} = p_1x_1 + p_2x_2 + p_3x_3 + p_4x_4 + \ldots + p_nx_n$$

Example: A single fair six-sided die is rolled.

Outcome	1	2	3	4	5	6
Probability	1/6	1/6	1/6	1/6	1/6	1/6

$$\text{Expected Value} = 1\cdot\frac{1}{6} + 2\cdot\frac{1}{6} + 3\cdot\frac{1}{6} + 4\cdot\frac{1}{6} + 5\cdot\frac{1}{6} + 6\cdot\frac{1}{6} = 3.5$$

21/6

In our previous tutorial where we talked about expected value's application for the prisoner's dilemma, we assume that the likelihood for the opponent's strategy is evenly spread (i.e., 50%). This assumption is what we call causal decision theory; in contrast to evidential decision theory. Let's look at some examples.

based on (caused) by your very action

based on evidence

for A player EV= 2 when he cooperate;

Set up: = 1 when he defect

	Player A	
Actions	Cooperate	Defect
Cooperate	(1, 1)	(3, 0)
Defect	(0, 3)	(2, 2)

Player B

Causal Decision Theory (CDT)

CDT evaluates each action based on the causal consequences directly brought about by the choice. The probability in CDT does not consider what the other player might do as influenced by one's own choice, but rather what outcome each action directly leads to given the other player's independent choice.

For instance, if Player A believes there is a 50% chance player B will cooperate, then her expected utility for defecting or cooperating is calculated as follows:

1. Defect: $0.5\times0+0.5\times2=1$ years in prison on average.
2. Cooperate: $0.5\times1+0.5\times3=2.5$ years in prison on average.

According to CDT, Player A should defect because defecting leads to a lower expected sentence, regardless of Player B’s decision.

Evidential Decision Theory (EDT)
EDT advises selecting the action that gives the best indication (evidence) of the outcome. Here, the choice made by Alice would depend on what she believes Bob's action indicates about her own situation. Iterated prisoner's dilemma

If Alice thinks her decision to cooperate or defect somehow **evidentially correlates**[2] with Bob's decision:

1. If cooperating indicates that Bob will also cooperate, the outcome might look favorable.
2. If defecting indicates that Bob will defect, the outcome is less favorable.

The probability assessment under EDT might involve reasoning like:

Alice might reason that cooperating is evidence that Bob will cooperate because they are both rational and would both prefer the lower combined sentence of cooperating (2 years total) over the higher combined sentence of defecting (4 years total). Thus, if A and B both believe that cooperation is a signal of mutual cooperation, they might both decide to cooperate.

this belief does not come from causal analysis
but it comes from our 'experiences'

[2] In econometric/statistics, evidential correlation is 'concluded' or derived from empirical data.